

Hazards_Unit

Function

Forward the data in case of data hazards and stall one cycle if there is a data hazard due to a memory instruction. If any instruction is using memory, it stalls fetching.

If the execute uses ra or rb and the instruction in the memory or wb destination is one of them, fwd_memory_execute of fwd_wb_execute is high respectively, and in case of that the instruction in the memory uses the memory, fetching is stalled by stall_structural and if its destination is used by the one in execute the instruction in the execute waits there waiting for data from memory by stall_data.

Inputs

- execute_use_ra
- execute_use_rb
- memory_writes
- wb_writes
- is_memory_instruction
- execute_ra[1:0]
- execute_rb[1:0]
- mem_dist[1:0]
- wb_dist[1:0]

Outputs

- fwd_A_memory_execute
- fwd_B_memory_execute
- fwd_A_wb_execute
- fwd_B_wb_execute
- stall_structural
- stall_data